Python Basics: RAG Chatbot Test Document

Small test PDF containing 10 questions and reference answers

Purpose: Use this document to test PDF upload, question answering, and answer evaluation in your RAG chatbot.

Question 1: What is Python?

Reference answer: Python is a high-level, interpreted programming language known for its readable syntax. It is commonly used for web development, automation, data analysis, and artificial intelligence.

Question 2: What is a variable in Python?

Reference answer: A variable is a name that refers to a value. For example, name = 'Sam' assigns the string 'Sam' to the variable name.

Question 3: What is a Python list?

Reference answer: A list is an ordered, mutable collection in Python. It can contain multiple items, which can be accessed by index. Example: numbers = [1, 2, 3].

Question 4: What is the difference between a list and a tuple in Python?

Reference answer: A list is mutable, so its contents can be changed after creation. A tuple is immutable, so its contents cannot be changed after creation.

Question 5: What is a Python function?

Reference answer: A function is a reusable block of code that performs a task. It is defined using the def keyword and can accept parameters and return a result.

Question 6: What does a dictionary store in Python?

Reference answer: A dictionary stores data as key-value pairs. Each key maps to a value. Example: student = {'name': 'Ravi', 'age': 21}.

Question 7: What is a Python loop used for?

Reference answer: A loop repeats a block of code. Python provides for loops to iterate over items and while loops to repeat while a condition remains true.

Question 8: What is exception handling in Python?

Reference answer: Exception handling lets a program respond to runtime errors. Python commonly uses try and except blocks to handle exceptions.

Question 9: What is a Python module?

Reference answer: A module is a Python file containing reusable code, such as functions, classes, and variables, that can be imported into another Python program.

Question 10: What is the purpose of a return statement?

Reference answer: A return statement ends a function call and sends a result back to the code that called the function. Without an explicit return statement, a Python function returns None.

Testing note: The chatbot may paraphrase the reference answers. Evaluate factual meaning, not exact wording alone.